

2.

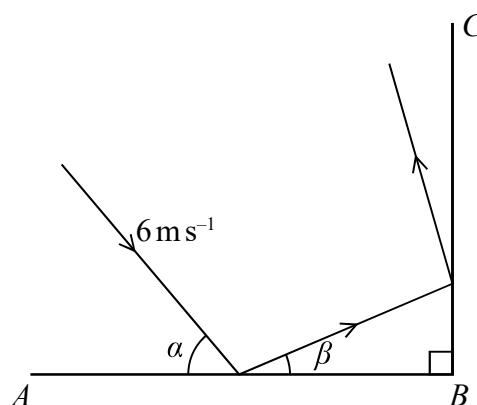


Figure 2

Figure 2 represents the plan view of part of a horizontal floor, where AB and BC are fixed vertical walls with AB perpendicular to BC .

A small ball is projected along the floor towards AB with speed 6 m s^{-1} on a path that makes an angle α with AB , where $\tan \alpha = \frac{4}{3}$. The ball hits AB and then hits BC .

Immediately after hitting AB , the ball is moving at an angle β to AB , where $\tan \beta = \frac{1}{3}$.

The coefficient of restitution between the ball and AB is e .

The coefficient of restitution between the ball and BC is $\frac{1}{2}$.

By modelling the ball as a particle and the floor and walls as being smooth,

- (a) show that the value of $e = \frac{1}{4}$ (5)
- (b) find the speed of the ball immediately after it hits BC . (4)
- (c) Suggest two ways in which the model could be refined to make it more realistic. (2)



4. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors in a horizontal plane.]

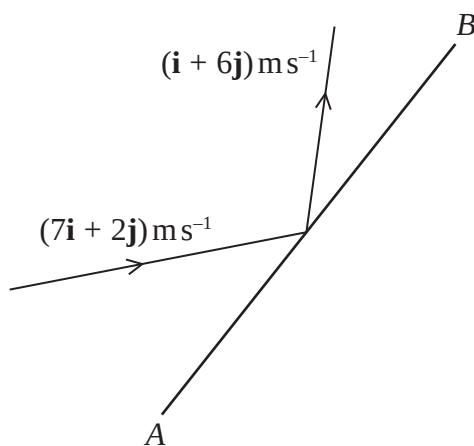


Figure 1

Figure 1 represents the plan view of part of a smooth horizontal floor, where AB represents a fixed smooth vertical wall.

A small ball of mass 0.5 kg is moving on the floor when it strikes the wall.

Immediately before the impact the velocity of the ball is $(7\mathbf{i} + 2\mathbf{j})\text{ m s}^{-1}$.

Immediately after the impact the velocity of the ball is $(\mathbf{i} + 6\mathbf{j})\text{ m s}^{-1}$.

The coefficient of restitution between the ball and the wall is e .

- (a) Show that AB is parallel to $(2\mathbf{i} + 3\mathbf{j})$.

(4)

- (b) Find the value of e .

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



4.

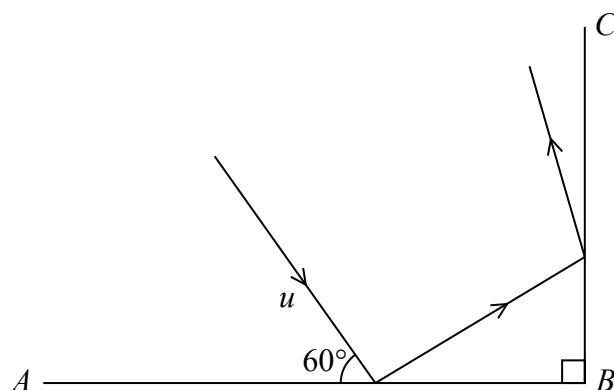
**Figure 1**

Figure 1 represents the plan view of part of a horizontal floor, where AB and BC are perpendicular vertical walls.

The floor and the walls are modelled as smooth.

A ball is projected along the floor towards AB with speed $u \text{ m s}^{-1}$ on a path at an angle of 60° to AB . The ball hits AB and then hits BC .

The ball is modelled as a particle.

The coefficient of restitution between the ball and wall AB is $\frac{1}{\sqrt{3}}$

The coefficient of restitution between the ball and wall BC is $\sqrt{\frac{2}{5}}$

- (a) Show that, using this model, the final kinetic energy of the ball is 35% of the initial kinetic energy of the ball.

(8)

- (b) In reality the floor and the walls may not be smooth. What effect will the model have had on the calculation of the percentage of kinetic energy remaining?

(1)

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors in a horizontal plane.]

A smooth uniform sphere A has mass $2m$ kg and another smooth uniform sphere B , with the same radius as A , has mass $3m$ kg.

The spheres are moving on a smooth horizontal plane when they collide obliquely.

Immediately before the collision the velocity of A is $(3\mathbf{i} + 3\mathbf{j})\text{ms}^{-1}$ and the velocity of B is $(-5\mathbf{i} + 2\mathbf{j})\text{ms}^{-1}$.

At the instant of collision, the line joining the centres of the spheres is parallel to $\mathbf{i}$.

The coefficient of restitution between the spheres is $\frac{1}{4}$

(a) Find the velocity of B immediately after the collision.

(7)

(b) Find, to the nearest degree, the size of the angle through which the direction of motion of B is deflected as a result of the collision.

(2)

7. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors in a horizontal plane.]

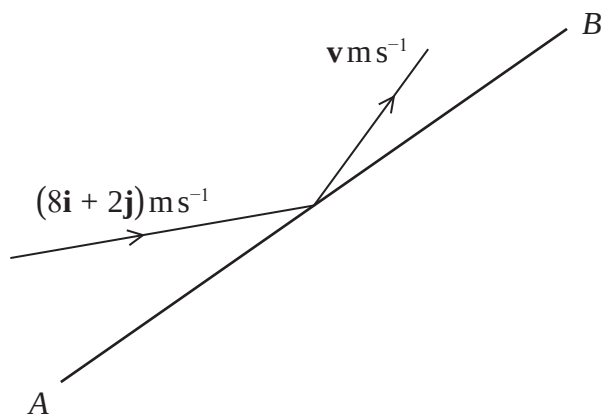


Figure 3

Figure 3 represents the plan view of part of a smooth horizontal floor, where AB is a fixed smooth vertical wall.

The direction of $\overrightarrow{AB}$ is in the direction of the vector $(\mathbf{i} + \mathbf{j})$

A small ball of mass 0.25 kg is moving on the floor when it strikes the wall AB .

Immediately before its impact with the wall AB , the velocity of the ball is $(8\mathbf{i} + 2\mathbf{j}) \text{ ms}^{-1}$

Immediately after its impact with the wall AB , the velocity of the ball is $\mathbf{v} \text{ ms}^{-1}$

The coefficient of restitution between the ball and the wall is $\frac{1}{3}$

By modelling the ball as a particle,

(a) show that $\mathbf{v} = 4\mathbf{i} + 6\mathbf{j}$ (6)

(b) Find the magnitude of the impulse received by the ball in the impact. (3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



5.

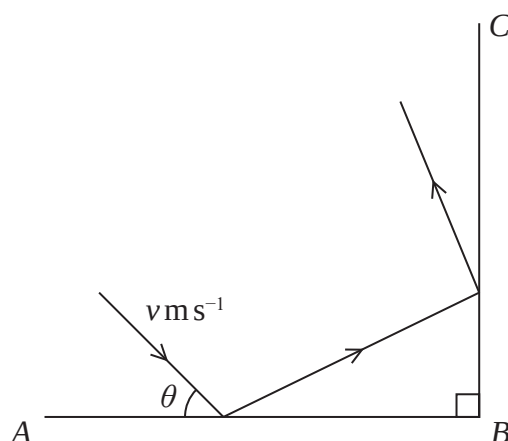
**Figure 1**

Figure 1 represents the plan view of part of a horizontal floor, where AB and BC represent fixed vertical walls, with AB perpendicular to BC .

A small ball is projected along the floor towards the wall AB . Immediately before hitting the wall AB the ball is moving with speed $v \text{ ms}^{-1}$ at an angle θ to AB .

The ball hits the wall AB and then hits the wall BC .

The coefficient of restitution between the ball and the wall AB is $\frac{1}{3}$.

The coefficient of restitution between the ball and the wall BC is e .

The floor and the walls are modelled as being smooth.

The ball is modelled as a particle.

The ball loses half of its kinetic energy in the impact with the wall AB .

(a) Find the exact value of $\cos \theta$.

(5)

The ball loses half of its remaining kinetic energy in the impact with the wall BC .

(b) Find the exact value of e .

(5)



6. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

A particle P is moving with velocity $(4\mathbf{i} - \mathbf{j})\text{ m s}^{-1}$ on a smooth horizontal plane.
The particle collides with a smooth vertical wall and rebounds with velocity $(\mathbf{i} + 3\mathbf{j})\text{ m s}^{-1}$

The coefficient of restitution between P and the wall is e .

(a) Find the value of e .

(6)

After the collision, P goes on to hit a second smooth vertical wall, which is parallel to $\mathbf{i}$.

The coefficient of restitution between P and this second wall is $\frac{1}{3}$

The angle through which the direction of motion of P has been deflected by its collision with this second wall is α° .

(b) Find the value of α , giving your answer to the nearest whole number.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors in a horizontal plane.]

A smooth uniform sphere A has mass 0.2 kg and another smooth uniform sphere B , with the same radius as A , has mass 0.4 kg .

The spheres are moving on a smooth horizontal surface when they collide obliquely. Immediately before the collision, the velocity of A is $(3\mathbf{i} + 2\mathbf{j})\text{ m s}^{-1}$ and the velocity of B is $(-4\mathbf{i} - \mathbf{j})\text{ m s}^{-1}$

At the instant of collision, the line joining the centres of the spheres is parallel to $\mathbf{i}$

The coefficient of restitution between the spheres is $\frac{3}{7}$

(a) Find the velocity of A immediately after the collision.

(7)

(b) Find the magnitude of the impulse received by A in the collision.

(2)

(c) Find, to the nearest degree, the size of the angle through which the direction of motion of A is deflected as a result of the collision.

(3)



4.

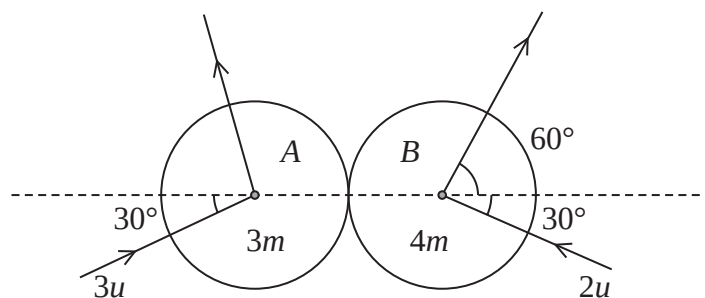


Figure 3

Two smooth uniform spheres, A and B , have equal radii. The mass of A is $3m$ and the mass of B is $4m$. The spheres are moving on a smooth horizontal plane when they collide obliquely. Immediately before they collide, A is moving with speed $3u$ at 30° to the line of centres of the spheres and B is moving with speed $2u$ at 30° to the line of centres of the spheres. The direction of motion of B is turned through an angle of 90° by the collision, as shown in Figure 3.

- Find the size of the angle through which the direction of motion of A is turned as a result of the collision.
- Find, in terms of m and u , the magnitude of the impulse received by B in the collision.

(9)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



3. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors in a horizontal plane.]

A smooth uniform sphere P has mass 0.3 kg . Another smooth uniform sphere Q , with the same radius as P , has mass 0.5 kg .

The spheres are moving on a smooth horizontal surface when they collide obliquely. Immediately before the collision the velocity of P is $(u\mathbf{i} + 2\mathbf{j})\text{ ms}^{-1}$, where u is a positive constant, and the velocity of Q is $(-4\mathbf{i} + 3\mathbf{j})\text{ ms}^{-1}$

At the instant when the spheres collide, the line joining their centres is parallel to $\mathbf{i}$.

The coefficient of restitution between P and Q is $\frac{3}{5}$

As a result of the collision, the direction of motion of P is deflected through an angle of 90° and the direction of motion of Q is deflected through an angle of α°

- (a) Find the value of u (8)
- (b) Find the value of α (5)
- (c) State how you have used the fact that P and Q have equal radii. (1)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



5. A smooth uniform sphere P has mass 0.3 kg . Another smooth uniform sphere Q , with the same radius as P , has mass 0.2 kg .

The spheres are moving on a smooth horizontal surface when they collide obliquely. Immediately before the collision the velocity of P is $(4\mathbf{i} + 2\mathbf{j})\text{ m s}^{-1}$ and the velocity of Q is $(-3\mathbf{i} + \mathbf{j})\text{ m s}^{-1}$.

At the instant of collision, the line joining the centres of the spheres is parallel to $\mathbf{i}$.

The kinetic energy of Q immediately after the collision is half the kinetic energy of Q immediately before the collision.

(a) Find

- (i) the velocity of P immediately after the collision,
 - (ii) the velocity of Q immediately after the collision,
 - (iii) the coefficient of restitution between P and Q ,
- carefully justifying your answers.

(11)

- (b) Find the size of the angle through which the direction of motion of P is deflected by the collision.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



8.

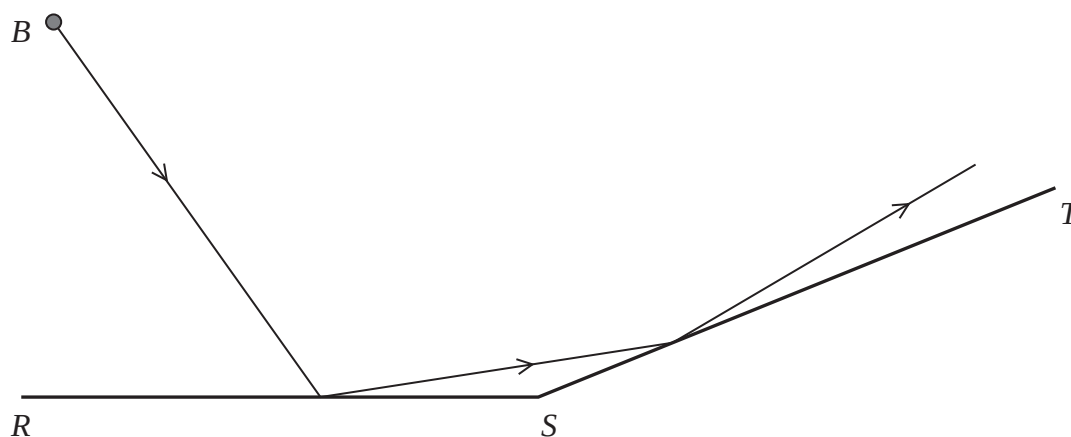


Figure 5

Figure 5 represents the plan view of part of a smooth horizontal floor, where RS and ST are smooth fixed vertical walls. The vector $\overrightarrow{RS}$ is in the direction of $\mathbf{i}$ and the vector $\overrightarrow{ST}$ is in the direction of $(2\mathbf{i} + \mathbf{j})$.

A small ball B is projected across the floor towards RS . Immediately before the impact with RS , the velocity of B is $(6\mathbf{i} - 8\mathbf{j})\text{ m s}^{-1}$. The ball bounces off RS and then hits ST .

The ball is modelled as a particle.

Given that the coefficient of restitution between B and RS is e ,

(a) find the full range of possible values of e .

(3)

It is now given that $e = \frac{1}{4}$ and that the coefficient of restitution between B and ST is $\frac{1}{2}$

(b) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the velocity of B immediately after its impact with ST .

(7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



7.

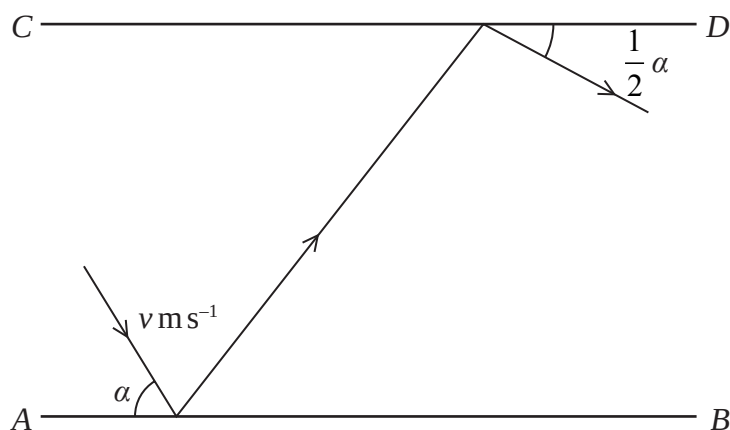
**Figure 2**

Figure 2 represents the plan view of part of a horizontal floor, where AB and CD represent fixed vertical walls, with AB parallel to CD .

A small ball is projected along the floor towards wall AB . Immediately before hitting wall AB , the ball is moving with speed $v \text{ m s}^{-1}$ at an angle α to AB , where $0 < \alpha < \frac{\pi}{2}$

The ball hits wall AB and then hits wall CD .

After the impact with wall CD , the ball is moving at angle $\frac{1}{2}\alpha$ to CD .

The coefficient of restitution between the ball and wall AB is $\frac{2}{3}$

The coefficient of restitution between the ball and wall CD is also $\frac{2}{3}$

The floor and the walls are modelled as being smooth. The ball is modelled as a particle.

(a) Show that $\tan\left(\frac{1}{2}\alpha\right) = \frac{1}{3}$ (7)

(b) Find the percentage of the initial kinetic energy of the ball that is lost as a result of the two impacts. (4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



6. A particle P of mass m is falling vertically when it strikes a fixed smooth inclined plane. The plane is inclined to the horizontal at an angle α , where $0 < \alpha \leq 45^\circ$

At the instant immediately before the impact, the speed of P is u .

At the instant immediately after the impact, P is moving horizontally with speed v .

- (a) Show that the magnitude of the impulse exerted on the plane by P is $mu \sec \alpha$ (5)

The coefficient of restitution between P and the plane is e , where $e > 0$

- (b) Show that $v^2 = u^2(\sin^2 \alpha + e^2 \cos^2 \alpha)$ (3)

- (c) Show that the kinetic energy lost by P in the impact is

$$\frac{1}{2}mu^2(1 - e^2)\cos^2 \alpha$$

(2)

- (d) Hence find, in terms of m , u and e **only**, the kinetic energy lost by P in the impact. (2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



7.

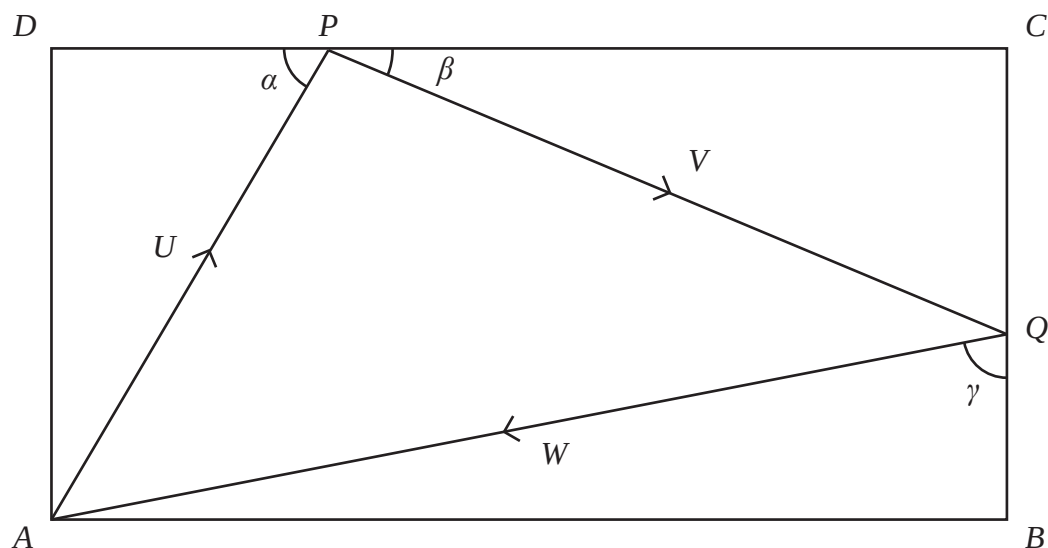


Figure 2

A small smooth snooker ball is projected from the corner A of a horizontal rectangular snooker table $ABCD$.

The ball is projected so it first hits the side DC at the point P , then hits the side CB at the point Q and then returns to A .

Angle $APD = \alpha$, Angle $QPC = \beta$, Angle $AQB = \gamma$

The ball moves along AP with speed U , along PQ with speed V and along QA with speed W , as shown in Figure 2.

The coefficient of restitution between the ball and side DC is e_1

The coefficient of restitution between the ball and side CB is e_2

The ball is modelled as a particle.

Use the model to answer all parts of this question.

- (a) Show that $\tan \beta = e_1 \tan \alpha$ (4)
- (b) Hence show that $e_1 \tan \alpha = e_2 \cot \gamma$ (3)
- (c) By considering (angle APQ + angle AQP) or otherwise, show that it would be possible for the ball to return to A only if $e_2 > e_1$ (6)

If instead $e_1 = e_2$, the ball would **not** return to A .

Given that $e_1 = e_2$

- (d) use the result from part (b) to describe the path of the ball after it hits CB at Q , explaining your answer. (1)



7.

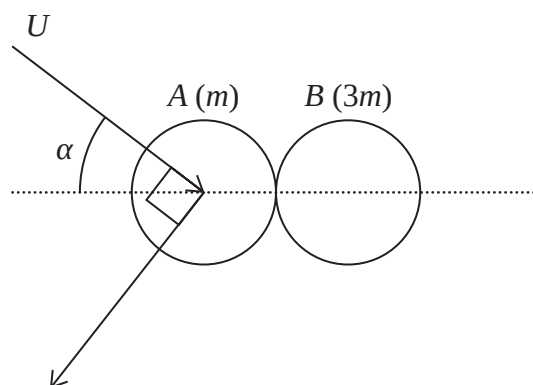


Figure 1

A smooth uniform sphere A of mass m is moving with speed U on a smooth horizontal plane. The sphere A collides obliquely with a smooth uniform sphere B of mass $3m$ which is at rest on the plane. The two spheres have the same radius.

Immediately before the collision, the direction of motion of A makes an angle α , where $0^\circ < \alpha < 90^\circ$, with the line joining the centres of the spheres.

Immediately after the collision, the direction of motion of A is **perpendicular** to its original direction, as shown in Figure 1.

The coefficient of restitution between the spheres is e .

(a) Show that the speed of B immediately after the collision is

$$\frac{1}{4}(1+e)U\cos\alpha \quad (6)$$

(b) Show that $e > \frac{1}{3}$ (4)

(c) Show that $0 < \tan\alpha \leq \frac{1}{\sqrt{2}}$ (5)

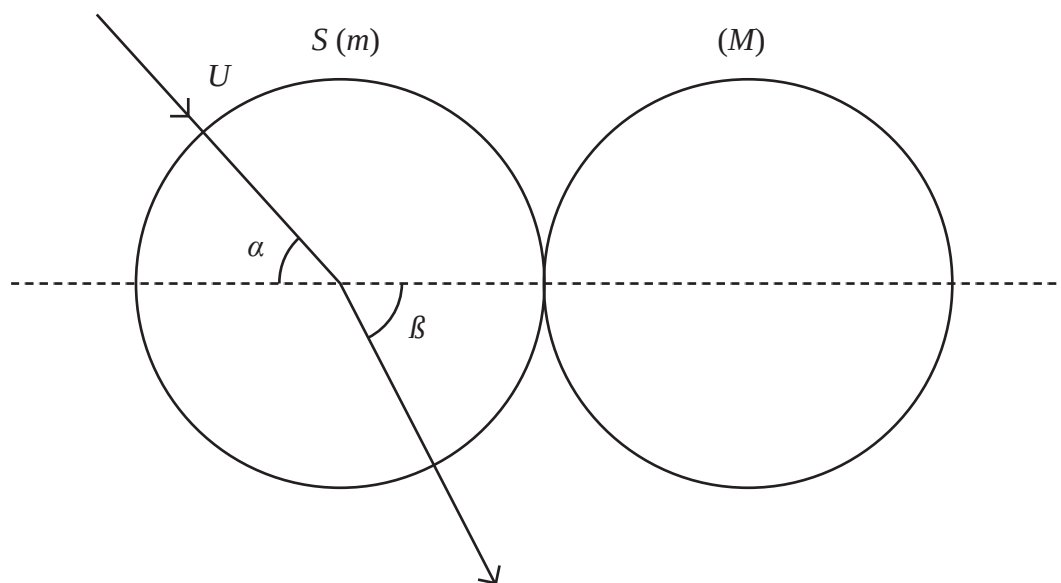
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



5.

**Figure 1**

A smooth uniform sphere S of mass m is moving with speed U on a smooth horizontal plane. The sphere S collides obliquely with another uniform sphere of mass M which is at rest on the plane. The two spheres have the same radius.

Immediately before the collision the direction of motion of S makes an angle α , where $0 < \alpha < 90^\circ$, with the line joining the centres of the spheres.

Immediately after the collision the direction of motion of S makes an angle β with the line joining the centres of the spheres, as shown in Figure 1.

The coefficient of restitution between the spheres is e .

(a) Show that $\tan \beta = \frac{(m + M) \tan \alpha}{(m - eM)}$ (8)

Given that $m = eM$,

(b) show that the directions of motion of the two spheres immediately after the collision are perpendicular. (2)

